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What is Third Normal Form (3NF)?

Definition

A relation is in Third Normal Form (3NF) if:

- It is already in Second Normal Form (2NF).
- It contains no Transitive Dependencies.

Formal Definition

For every Functional Dependency:

```
X → A
```

At least one must be true:

- X is a Super Key, OR
- A is a Prime Attribute (part of some Candidate Key).

Converting to 3NF

Original

EmployeeID	EmployeeName	DepartmentID	DepartmentName

FDs

```
EmployeeID → EmployeeName
EmployeeID → DepartmentID
DepartmentID → DepartmentName
```

Step 1

Identify the transitive dependency.

```
DepartmentID → DepartmentName
```

Step 2

Create a separate Department table.

Employee

EmployeeID	EmployeeName	DepartmentID

Department

DepartmentID	DepartmentName

Now each fact is stored only once.

Dependency Preservation

Definition

A decomposition is Dependency Preserving if all original Functional Dependencies can still be enforced without joining tables.

If dependencies are not preserved:

- More joins are needed to validate data.
- Updates become more expensive.

3NF allows one exception:
A dependency is acceptable if the dependent attribute is a Prime Attribute.
Sometimes this still leaves redundancy.
BCNF removes this exception.

BCNF Rule

A relation is in BCNF if:

```
Every determinant is a Candidate Key.
```

Why Isn't 3NF Always Enough?

Consider

Teacher	Course	Room

FDs

```
Teacher → Room
Course → Teacher
```

Suppose each teacher has one room.
Each course has one teacher.

Candidate Keys

```
Course
```

Since

> **`Teacher → Room`**

Teacher is not a Candidate Key.
The table violates BCNF.
Yet it can satisfy 3NF depending on the key structure and prime attributes.

```
Teacher
   |
   ▼
 Room


Course
   |
   ▼
Teacher
```

Teacher determines Room but is not a Candidate Key.
Violation.

BCNF Algorithm

1. Find Functional Dependencies.
2. Find Candidate Keys.
3. Find any FD where the determinant is not a Candidate Key.
4. Decompose the relation.
5. Repeat until every relation satisfies BCNF.

BCNF sometimes lose Dependency Preservation?

Answer

BCNF focuses on removing all violations where a determinant is not a Candidate Key.
During decomposition, some original Functional Dependencies may become split across multiple tables. Checking or enforcing those dependencies may require joining tables, so the decomposition is not dependency preserving.

> ***3NF*** *is often preferred when preserving dependencies is important.*
>
> ***BCNF*** *is preferred when eliminating redundancy is more important.*